

Adherence to Behavioral Interventions for Urge Incontinence When Combined With Drug Therapy: Adherence Rates, Barriers, and Predictors

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Background. Behavioral intervention outcomes for urinary incontinence (UI) depend on active patient participation.

Objective. The purpose of this study was to describe adherence to behavioral interventions (pelvic-floor muscle [PFM] exercises, UI prevention strategies, and delayed voiding), patient-perceived exercise barriers, and predictors of exercise adherence in women with urge-predominant UI.

Design. This was a prospectively planned secondary data analysis from a 2-stage, multicenter, randomized clinical trial.

Patients and Intervention. Three hundred seven women with urge-predominant UI were randomly assigned to receive either 10 weeks of drug therapy only or 10 weeks of drug therapy combined with a behavioral intervention for UI. One hundred fifty-four participants who received the combined intervention were included in this analysis.

Measurements. Pelvic-floor muscle exercise adherence and exercise barriers were assessed during the intervention phase and 1 year afterward. Adherence to UI prevention strategies and delayed voiding were assessed during the intervention only.

Results. During intervention, 81% of women exercised at least 5 to 6 days per week, and 87% performed at least 30 PFM contractions per day. Ninety-two percent of the women used the urge suppression strategy successfully. At the 12-month follow-up, only 32% of the women exercised at least 5 to 6 days per week, and 56% performed 15 or more PFM contractions on the days they exercised. The most persistent PFM exercise barriers were difficulty remembering to exercise and finding time to exercise. Similarly, difficulty finding time to exercise persisted as a predictor of PFM exercise adherence over time.

Limitations. Co-administration of medication for UI may have influenced adherence.

Conclusions. Most women adhered to exercise during supervised intervention; however, adherence declined over the long term. Interventions to help women remember to exercise and to integrate PFM exercises and UI prevention strategies into daily life may be useful to promote long-term adherence.

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Urinary Incontinence Treatment Network (see list of investigators and participants in Appendix 1).

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Urge urinary incontinence (UI), defined as the involuntary loss of urine associated with a strong desire to void, is a common problem among women, particularly older women.^{1,2} First-line interventions for UI include pharmacotherapy and behavioral interventions.^{3,4} Bladder training,^{5,6} pelvic-floor muscle (PFM) exercise,⁷ and urge suppression strategy training^{8,9} are common components of a UI behavioral intervention program. Positive outcomes of these interventions likely depend on patient adherence to the prescribed regimen.

Reasons women give for poor PFM exercise adherence include forgetting to exercise; lacking time, interest, or discipline to exercise; and interference of exercises with daily activities.^{10,11} The amount of urine loss per incontinence episode and a woman's belief in her ability to exercise have been identified as important predictors of a woman's intent to adhere to PFM exercise.¹² For long-term PFM exercise adherence, short-term adherence and urinary incontinence (UI) episode frequency have been found extremely important.¹³ In women postpartum, higher parity, education, and general fitness levels and greater UI symptoms were found to predict regular adherence to PFM exercise 6 months postpartum.¹⁴

Although long-term data are sparse, continued PFM exercise is associated with better long-term outcomes. In

logistic regression analysis with age, parity, health locus of control, and incontinence severity, continued PFM exercise was the only factor significantly associated with better 5-year posttherapy outcomes.¹⁵ As women tend to decrease PFM exercise frequency over time,^{10,15-17} a greater understanding of PFM exercise adherence is needed. Studies of PFM exercise adherence have focused mainly on women postpartum or women with stress UI.^{10,11,14,16,17} Additionally, the impact of exercise barriers on PFM exercise adherence has not been well-studied.¹²⁻¹⁴

This article reports on a prospectively planned secondary analysis of PFM exercise and behavioral strategy adherence data collected during the Behavior Enhances Drug Reduction of Incontinence (BE-DRI) study, a multi-site, randomized controlled trial comparing drug therapy alone with combined behavioral and drug therapy for urge-predominant UI.¹⁸ The purposes of the present study were: (1) to describe adherence to PFM exercises and UI prevention strategies; (2) to describe barriers to PFM exercise, as reported by participants; and (3) to identify factors, including exercise barriers, associated with PFM exercise adherence.

Method

Design Overview

The BE-DRI study was a 2-stage randomized clinical trial conducted by 9 clinical centers in the United States (see Appendix 1 for participating centers). Participants were women with UI who were randomly assigned to receive either 10 weeks of drug therapy only or 10 weeks of drug therapy in combination with a behavioral intervention for UI. Details of the study design and method have been described previously.¹⁹ This article reports adherence data from the subgroup of women who received the behavioral intervention in combination with drug therapy. In

brief, we describe the findings from the clinical trial. At 10 weeks, 69% of women in the combined study arm achieved a $\geq 70\%$ reduction in UI episodes recorded on a bladder diary. At 10 weeks, drug therapy was discontinued, but the women were advised to continue their behavioral intervention program. Eight months following the start of treatment, based on life-table analysis, 41% of the women achieved treatment success, defined as remaining off drug therapy and having a $\geq 70\%$ reduction in frequency of UI episodes.¹⁸ All participants provided written informed consent.

Setting and Participants

Participants were community-dwelling women recruited through the investigators' clinical practices, study announcements, advertisements, and referrals. Only women with pure or predominant UI, defined as urge symptom index $>$ stress symptom index on the Medical, Epidemiological, and Social Aspects of Aging Questionnaire (MESA),²⁰ were considered for clinical evaluation. To be eligible for the study, women had to demonstrate a post-void residual volume of less than 150 mL and the ability to contract their PFMs. They also had to show 7 or more episodes of UI on a 7-day baseline diary. Finally, they had to self-report persistent UI for at least 3 months, no current use of antimuscarinics or other medications that could affect UI, and no history of neurologic diseases or conditions (eg, Parkinson disease, multiple sclerosis, spina bifida, spinal cord injury) or systemic diseases known to affect bladder function.

Intervention Phase

The 10-week intervention phase included 4 visits at intervals of 2 to 3 weeks. All women completed a daily bladder diary throughout this phase. Drug therapy was 4 mg per day of open-label tolterodine tartrate



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(extended-release capsules). At the first visit, participants received a 30-day drug supply, drug information, and recommendations for fluid management. At subsequent visits, the dose could be decreased to 2 mg to minimize side effects, or if not tolerated, another antimuscarinic medication could be substituted.

The behavioral intervention was implemented by a trained physical therapist, nurse practitioner, or nurse specialist. To ensure standardized administration across clinicians and clinical sites, behavioral interventionists attended centralized training. They were certified on patient education skills related to PFM exercise, UI prevention strategies, fluid management, and delayed voiding. Certification also required interventionists to provide strategies to minimize exercise barriers. For example, possible solutions to “forgetting to exercise” include posting a reminder note on a bathroom mirror or computer screen, setting aside a consistent time for exercise each day, and associating exercise with another established behavior such as brushing one’s teeth.

Other means to standardize the behavioral intervention included use of encounter forms that required the interventionists to check off each element of the behavioral intervention as it was completed and standardized patient education handouts. The interventionists attended bi-monthly telephone conferences conducted by the study co-investigators to review study procedures and to discuss active cases and any problems that arose during the trial.

The behavioral intervention included PFM exercises; strategies to diminish urgency, suppress bladder contractions, and prevent urge incontinence^{8,9}; strategies to prevent stress incontinence for those with mixed UI symptoms^{8,9}; delayed void-

ing to increase voiding intervals (gradually by 10-minute increments until voiding every 3–4 hours during the day) for those who voided 8 or more times per day and were able to successfully suppress an urge while walking to the toilet; and individualized fluid management for those with excessive urine output (>70 oz [2,070.15 mL] per day).

Throughout the intervention phase, the women were prescribed a maximum of 45 PFM contractions per day (15 PFM contractions, 3 times per day). The protocol allowed prescription of fewer PFM contractions for women with very weak muscles. The maximum prescribed PFM contraction/relaxation duration progressed throughout the intervention phase from 3 seconds/6 seconds at visit 1 to a maximum of 10 seconds/20 seconds at visit 4.

After the end of 10 weeks, drug therapy was discontinued, but the women were advised to continue

their behavioral intervention program. They were provided with a maintenance exercise program of 15 PFM contractions per day, at the maximum contraction duration achieved during the intervention phase.

Adherence Outcomes

Pelvic-floor muscle exercise adherence was measured using 2 self-administered exercise questionnaires designed for this study. The questionnaires did not undergo psychometric testing. To establish content validity, we modeled the questionnaires after the one described by Sluijs et al.²¹ The intervention exercise questionnaire included 9 questions and was administered at visits 2 through 4. The first question asked women how often they had performed PFM exercises in the past week. Response choices were “every day,” “5 or 6 days,” “between 2 and 4 days,” and “no days.” The second question asked women to identify how many PFM contractions

The Bottom Line

What do we already know about this topic?

Data suggest that by following formalized therapy for urinary incontinence (UI), women who continue to perform pelvic-floor muscle (PFM) exercises report better long-term outcomes. Although some barriers to PFM exercise have been identified, little is known of their impact on exercise adherence.

What new information does this study offer?

In this study, 154 women with urge-predominant UI were advised on strategies to reduce barriers to PFM exercise. Exercise adherence rates were high during supervised intervention but declined over the 1-year follow-up period. Difficulty finding time to exercise persisted as a predictor of exercise adherence across time.

If you’re a patient, what might these findings mean for you?

Cost-effective methods for monitoring and promoting long-term PFM exercise in women with UI need to be identified and tested for efficacy.

they had performed on a typical day in the past week. Response choices were “all 45 contractions recommended by the interventionist,” “at least 30 contractions,” “at least 15 contractions,” and “fewer than 15 contractions.” The remaining 7 questions queried women about exercise barriers. Women were asked to respond “yes,” “no,” or “uncertain” to each of the questions (Appendix 2). At each visit, the interventionists reviewed and discussed each participant’s questionnaire responses to identify strategies to enhance adherence.

At visits 3 and 4, the interventionists also used a checklist to query women about whether they had implemented the urge suppression strategy (contracting the PFMs to suppress bladder contractions and prevent UI), the stress strategy (contracting the PFMs to occlude the urethra during a cough, sneeze, or other activities that result in stress UI), and delayed voiding (for those with frequent voiding).^{8,9} They also were asked whether the particular strategy was successful in reducing UI.

The exercise questionnaire used to obtain adherence data during the year following discontinuation of active therapy (at 2, 4, 6, and 12 months) was similar to that used during the intervention phase, except response options for the number of exercises performed were changed to: “more than 15 contractions,” “exactly 15 contractions,” or “fewer than 15 contractions.” Most of the exercise barrier questions were identical to those in the intervention exercise questionnaire (Appendix 2). Women did not meet with the interventionists or receive any additional exercise instruction during this follow-up period.

Data Analysis

Adherence to the behavioral interventions and barriers to exercise adherence were characterized using

descriptive statistics. The mean number of exercises per day was calculated as: (number of days exercised per week $\times$ the number of contractions per day) $\div$ 7. Because the exercise adherence responses were intervals, we used the midpoint of the response categories to calculate the mean. We then averaged these means over the entire intervention period and follow-up period separately. Exercise adherence during the intervention phase was defined as the mean number of exercises per day averaged at visits 2 through 4. For the purposes of calculating the mean, the response of fewer than 15 exercises per day was entered as 8 to account for the range of possibilities. Thus, if a woman exercised between 2 and 4 days per week (midpoint=3 days) and performed at least 15 contractions on a typical day (midpoint=22), we determined she performed 66 contractions that particular week (or 9.4 contractions per day).

Exercise adherence during the 1-year follow-up phase was defined as the mean number of exercises per day as reported at each of the 2, 4, 6 and 12-month visits. For the purpose of calculating mean adherence during follow-up, 25, 15, and 8 contractions were used for the 3 response categories (more than 15 contractions, 15 contractions, fewer than 15 contractions, respectively).

Associations of factors hypothesized to be related to exercise adherence were explored using least squares regression analysis. Factors to be tested were identified *a priori* and included: age, race, education level, baseline severity of UI (defined as mean number of accidents per day recorded in the bladder diary), MESA urge index,²⁰ MESA stress index,²⁰ baseline score on the Urogenital Distress Inventory,²² pretreatment fluid intake (from the bladder diary), PFM strength as determined via digital vaginal examination and quantified

using the Brink scale,²³ outcome expectations (“get very much better” or “get a little better or stay about the same”), 12-Item Short-Form Health Survey questionnaire (SF-12) score,²⁴ total number of reported barriers to exercise, and the response to each exercise barrier question. For analysis purposes, only a “yes” response to a barrier question was considered an endorsement of the exercise barrier. “Uncertain” and “no” responses were grouped together. Variables reaching a *P* value of $\leq .10$ in bivariate analyses then were entered into multivariable regression models to test their independent association with adherence. As data were obtained from multiple centers, study site was included as a random effect in the multivariable analyses. The same set of potential predictors was evaluated during treatment and follow-up, but the models were constructed separately. All analyses were carried out using SAS statistical software.*

Results

Between July 2004 and January 2006, 307 women were enrolled, and 154 were randomly assigned to receive the combined drug and behavioral treatment. One hundred forty-four women (94%) attended at least 3 therapy visits. Participant characteristics are presented in Table 1. Although all participants had UI, most (99%) also had symptoms of stress UI.

At the initial intervention visit, 150 women (97%) were able to contract their PFMs, and 149 women (97%) were able to perform 5 quick (1-second) contractions, with 138 (90%) able to do this without performing a Valsalva maneuver or contracting accessory muscles (ie, the gluteal or hip adductor muscles). By treatment visit 4 (*n*=143), 137 women (96%) were able to contract their PFMs without

* SAS Institute Inc, PO Box 8000, Cary, NC 27513.

performing a Valsalva maneuver or contracting accessory muscles.

Adherence to Behavioral Interventions

Figure 1 shows the change in exercise frequency over time during the intervention and follow-up periods. Initially, a high percentage of women were exercising daily. Exercise frequency decreased slightly over the intervention period; however, 116 women (81%) were still exercising 5 to 6 days or more per week at visit 4. However, by the end of the follow-up period (n=110), only 30 women (32%) were exercising at least 5 to 6 days per week, and the number of women who stopped exercising increased almost threefold.

Figures 2 and 3 illustrate the number of PFM contractions women performed during the intervention period and the follow-up period, respectively. Throughout the intervention period, most women reported doing at least 30 PFM contractions in a typical day. From the beginning to the end of the 1-year follow-up, the proportion of women who performed 15 or more contractions on the days they exercised decreased from two thirds to slightly above one half.

All women were taught the urge suppression strategy (ie, stay still and repeatedly contract the PFMs to control urgency and prevent UI). At visit 4 (n=143), 138 women (97%) reported that they had used the strategy, and 131 women (92%) indicated they successfully prevented a UI episode with this strategy. At visit 2 (n=147), 75 women (51%) were taught the stress strategy, as they indicated stress UI episodes in their bladder diary from the previous week. By visit 4 (n=143), 39 women (27%) reported using the strategy, and 35 women (25%) reported using it successfully to prevent stress UI. At visit 2, 97 women (66%) were

Table 1.

Characteristics of Participants in Combined Drug and Behavioral Treatment (n=154)

Characteristic	Values
Age (y), mean (SD)	55.8 (14.2)
Parous, n (%)	135 (88%)
Median no. of vaginal deliveries	2
Race/ethnicity, ^a n (%) (n=153)	
Hispanic	13 (8.5)
Non-Hispanic white	105 (69)
Non-Hispanic black	22 (14)
Non-Hispanic other	13 (8.5)
Marital status, n (%)	
Married/living as married	70 (45)
Not married	84 (55)
Level of education, n (%)	
High school or less	40 (26)
Some college/associate degree	58 (38)
Completed 4 years of college	32 (21)
Graduate/professional degree	24 (16)
Nam-Powers-Boyd Occupational Status Score, ^b mean (SD)	57.6 (23.3)
Duration of urge urinary incontinence (y), mean (SD)	9.8 (9.5)
Previous nonsurgical treatment for urinary incontinence, n (%)	59 (38)
Prior surgery for urinary incontinence, n (%)	19 (12)
Diuretic use, n (%)	15 (10)
Body mass index (kg/m ²), mean (SD)	33.2 (9.5)
Menopausal status, n (%)	
Premenopausal	35 (23)
Postmenopausal	96 (62)
Perimenopausal	20 (13)
Unsure	3 (2)
Randomization strata: type of incontinence and frequency of incontinence episodes per week, n (%)	
Urge, 7–13 episodes/wk	1 (0.7)
Urge, 14+ episodes/wk	1 (0.7)
Mixed, 7–13 episodes/wk	46 (29.9)
Mixed, 14+ episodes/wk	106 (68.8)

^a The race/ethnicity of participants was measured in this study to characterize the study sample. The response options were defined by the investigators, and participants classified themselves.

^b The Nam-Powers-Boyd Occupational Status Score ranks occupation based on educational requirements and expected salary (scale: 0–100, with higher score indicating greater status). Reference: Nam C, Boyd M. Occupational status in 2000: over a century of census-based measurement. *Popul Res Policy Rev*. 2004;23:327–358.

taught delayed voiding. By visit 4 (n=143), 84 women (59%) indicated they were able to delay voiding.

Barriers to Exercise Adherence

The percentages of women who endorsed each exercise barrier during the 10-week intervention and 1-year follow-up are shown in Tables 2 and

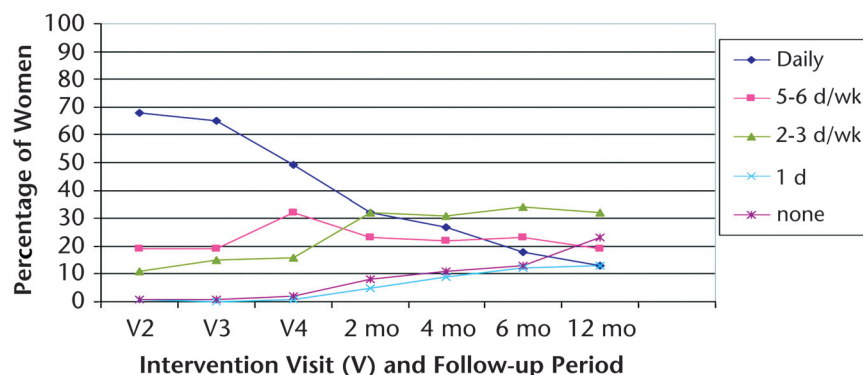


Figure 1.

Pelvic-floor muscle exercise frequency over time. Exercise frequency was high during the supervised intervention phase, with 81% of the women exercising 5 to 6 days or more per week at visit 4. Exercise frequency decreased throughout the follow-up phase, with only 32% of the women exercising at least 5 to 6 days per week.

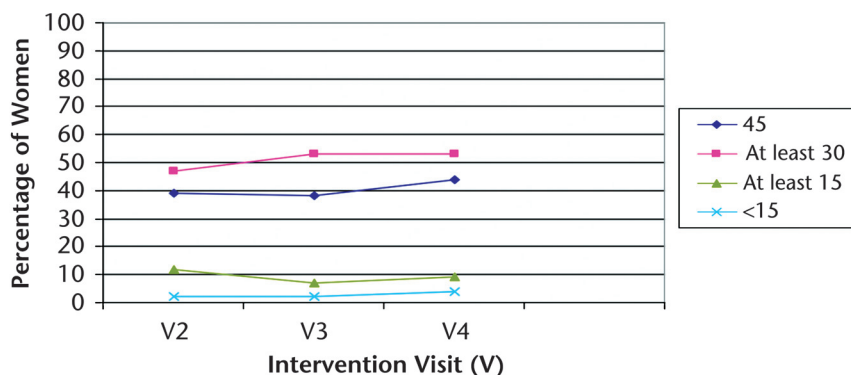


Figure 2.

Number of pelvic-floor muscle contractions per day over time during the intervention period. Throughout supervised intervention, about 90% of the women reported doing at least 30 pelvic-floor muscle contractions in a typical day.

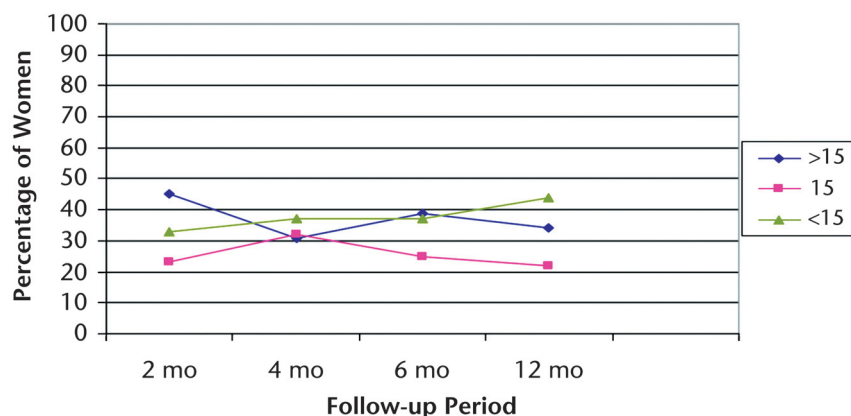


Figure 3.

Number of pelvic-floor muscle contractions per day over time during the follow-up period. By the 12-month follow-up, only slightly more than half of the women performed 15 or more pelvic-floor muscle contractions per day when they exercised.

3, respectively. “Other” barriers to exercise reported by the last treatment visit included illness, family matters, vacation or travel, too tired to exercise, and life stress. Similar “other” barriers were reported during the follow-up phase, with the addition of “I don’t like or I don’t want to exercise.”

Factors Associated With Exercise Adherence

In bivariate analyses, increasing age, the number of barriers reported, difficulty finding time to do all of the exercises, difficulty remembering to exercise, and an indication that “other reasons” were preventing exercise were found to be significantly associated ($P < .05$) with a lower number of PFM exercises performed during active intervention (Tab. 4). Concern that exercises were not helping incontinence was associated with higher exercise performance during the intervention phase. Increasing age and race approached but did not reach statistical significance ($P = .05$ and $P = .09$, respectively). None of the other individual barriers were associated with exercise adherence.

With a few exceptions, the results of the follow-up phase bivariate analyses were similar to the results of the active intervention phase (Tab. 4). In addition, higher baseline MESA stress index score and higher pretreatment fluid intake were significantly associated with exercise adherence ($P < .05$). Concern that exercises were not helping incontinence was not associated with exercise adherence. Less education approached but did not reach statistical significance ($P = .07$).

We computed the multivariable association of barriers with number of contractions controlling for age, race or ethnicity, MESA urge and stress indexes, and fluid intake pretreatment. The multivariable models were

Table 2.

Number (%) of Women Who Reported Exercise Barriers During the 10-Week Intervention

Barrier	Treatment Visit 2			Treatment Visit 3			Treatment Visit 4		
	Total n	Yes n (%)	Uncertain n (%)	Total n	Yes n (%)	Uncertain n (%)	Total n	Yes n (%)	Uncertain n (%)
Difficult to find time to do all of the exercises	146	45 (31%)	3 (2%)	139	59 (42%)	6 (4%)	143	53 (37%)	6 (4%)
Uncertain if performing exercises correctly	146	40 (27%)	8 (6%)	139	21 (15%)	4 (3%)	143	3 (2%)	10 (7%)
Exercise caused pain	146	7 (5%)	2 (1%)	139	3 (2%)	4 (3%)	143	8 (6%)	4 (3%)
Had difficulty remembering to exercise	146	59 (40%)	6 (4%)	139	42 (30%)	2 (1%)	143	49 (34%)	3 (2%)
Perceived exercises did not help	146	22 (15%)	59 (40%)	138	14 (10%)	26 (19%)	143	27 (19%)	31 (22%)
Improved urinary incontinence, no need to exercise	146	11 (8%)	16 (11%)	138	8 (6%)	7 (5%)	143	13 (9%)	8 (6%)
Medication more important than exercise	145	44 (30%)	31 (21%)	137	30 (22%)	35 (26%)	143	27 (19%)	22 (15%)
Other	143	15 (11%)	1 (1%)	135	16 (12%)	5 (4%)	141	24 (17%)	2 (1%)

conducted first with the total number of reported barriers and then with each barrier that was significant bivariate (Tab. 4). For multivariable analyses of the intervention period, the total number of barriers reported and the individual barriers remained

significantly associated with exercise adherence when the other participant characteristics were controlled. The slope coefficients indicate that a higher number of reported barriers or endorsement of individual barriers (with one exception) was associ-

ated with performance of fewer PFM contractions per day. For example, women who reported "difficulty finding time to do all the exercises" did an average of 8 fewer contractions per day compared with women who did not endorse this barrier.

Table 3.

Number (%) of Women Who Reported Exercise Barriers During the 1-Year Follow-Up

Barrier	2 Months			4 Months			6 Months			12 Months		
	Total n	Yes n (%)	Uncertain n (%)	Total n	Yes n (%)	Uncertain n (%)	Total n	Yes n (%)	Uncertain n (%)	Total n	Yes n (%)	Uncertain n (%)
Difficult to find time to do all of the exercises	121	37 (31%)	6 (5%)	107	45 (42%)	6 (6%)	110	40 (36%)	7 (6%)	97	41 (42%)	7 (7%)
Uncertain if performing exercises correctly	121	9 (7%)	6 (5%)	107	7 (7%)	4 (4%)	109	9 (8%)	8 (7%)	96	5 (5%)	6 (6%)
Exercise caused pain	121	5 (4%)	2 (2%)	107	1 (1%)	2 (2%)	110	6 (6%)	2 (2%)	95	2 (2%)	0 (0%)
Had difficulty remembering to exercise	121	48 (40%)	4 (3%)	108	48 (44%)	3 (3%)	110	58 (53%)	5 (5%)	96	54 (56%)	5 (5%)
Perceived exercises did not help	119	27 (23%)	20 (17%)	103	29 (28%)	16 (16%)	109	31 (28%)	24 (22%)	92	28 (30%)	24 (26%)
Improved urinary incontinence, no need to exercise	119	13 (11%)	5 (4%)	102	12 (12%)	4 (4%)	110	16 (15%)	5 (5%)	92	17 (19%)	8 (9%)
Other	111	18 (16%)	2 (2%)	100	14 (14%)	2 (2%)	101	16 (16%)	2 (2%)	89	6 (7%)	4 (5%)

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Table 4.

Association of Total and Individual Barriers With Exercise Adherence: Regression Coefficients, 95% Confidence Interval (CIs), and *P* Values

	Unadjusted ^a			Adjusted ^b		
	Estimate ^c	95% CI	<i>P</i>	Estimate ^c	95% CI	<i>P</i>
Intervention period						
Total no. of reported barriers	−1.7	−2.8, −0.6	.002	−2.0	−3.1, −0.9	.0007
Barrier: Difficult to find time to do all of the exercises	−6.4	−9.1, −3.7	<.001	−7.7	−11.1, −4.4	<.001
Barrier: Difficulty remembering to exercise	−4.8	−7.9, −1.1	.01	−7.5	−10.8, −4.2	<.001
Barrier: Perceived exercises do not help	4.1	0.4, 7.8	.03	4.2	0.4, 8.0	.03
Barrier: Other	−4.8	−8.5, −1.4	.006	−4.0	−8.1, −0.03	.048
Follow-up period						
Total no. of reported barriers	−0.9	−1.6, −0.1	.03	−0.6	−1.3, 0.2	.16
Barrier: Difficult to find time to do all of the exercises	−3.8	−6.1, −1.5	.001	−2.5	−4.7, −0.2	.03
Barrier: Difficulty remembering to exercise	−2.6	−4.9, −0.3	.03	−1.9	−4.2, 0.5	.12
Barrier: Perceived exercises do not help	2.0	−0.3, 4.4	.09	1.4	−1.0, 3.8	.24
Barrier: Other	−3.6	−5.8, −1.3	.002	−2.2	−4.8, 0.4	.10

^a Unadjusted: bivariate associations of total barriers and each individual barrier with number of pelvic-floor muscle contractions per day.

^b Adjusted: multivariable associations of total barriers and each individual barrier with number of pelvic-floor contractions per day, controlling for age; education; race/ethnicity; Medical, Epidemiological, and Social Aspects of Aging Questionnaire (MESA) urge index; MESA stress index; volume of fluid intake pretreatment; and clinical site.

^c Estimate: regression coefficient is the change in contractions per day per unit increase in total barriers or for endorsement of individual barrier versus no endorsement of that barrier.

However, endorsement of the barrier “exercises don’t help incontinence” was associated with a slightly higher performance of PFM contractions per day compared with nonendorsement of this barrier. None of the other participant characteristics were associated with exercise adherence when the barriers were controlled (data not shown).

In the follow-up phase, only the barrier “difficulty finding time to do all the exercises” remained significantly associated with a lower number of PFM exercises performed when the other participant characteristics were controlled (Tab. 4). The impact of this barrier on adherence was less than that seen in the intervention phase. Women who reported “difficulty finding time to do all the exercises” performed an average of 2.5

fewer PFM exercises per day compared with women who did not report this barrier. Among the other participant characteristics, higher fluid intake pretreatment remained significantly associated with a greater number of PFM exercises per day when the other factors were controlled (data not shown).

Discussion

Behavioral interventions, including PFM exercise, urge suppression strategies, stress strategies, and delayed voiding, are widely used interventions with established efficacy for stress and UUI. One of the limitations of behavioral interventions is that they rely on active patient participation. Motivating the patient to adhere to treatment recommendations and sustain her efforts over time is a central goal of this ap-

proach. This article provides important information on barriers to adherence that may help improve the implementation and outcomes of behavioral interventions for people with urge-predominant UI.

The results of this study indicate that during supervised intervention for female, urge-predominant UI, PFM exercise adherence is high. Eighty percent of the women exercised at least 5 times per week, and 90% performed at least 30 PFM contractions on the days they exercised. It is possible that the women in our study may have over-reported adherence in an effort to please the interventionists. To eliminate this bias, a “noninterventionist” could have been appointed to collect and review the exercise adherence questionnaires. However, in true clinical

practice, it is the clinician's role to monitor patient adherence to the prescribed exercise regimen.

When controlling for factors previously found to associate with adherence, such as age, educational status, and severity of incontinence, only the number of reported barriers and some individual barriers (difficulty finding time to do all the exercises, difficulty remembering to exercise, and "other reasons" for not exercising) were significantly associated with decreased PFM exercise adherence during active intervention. We believe our efforts to identify barriers throughout intervention promoted our high level of observed exercise adherence. Once barriers were identified, interventionists assisted participants in discovering strategies to minimize them. For example, if a woman reported difficulty finding time to exercise, the interventionist may have recommended a standard time of day to exercise. Alternately, she may have suggested integrating exercises with other well-established daily activities (washing dishes, sitting at stoplights during the drive to work, or watching the evening news).

At the 12-month follow-up, 64% of the women continued to exercise 2 or more days per week, but only 33% exercised 5 or more times per week. About half of the women performed at least 15 PFM contractions when they exercised. Our 12-month PFM exercise adherence rates were similar to or slightly higher than those reported in other studies. Chiarelli et al²⁵ reported only a third of women postpartum performed PFM exercises at least several times per week 12 months following completion of a supervised continence program. Hines et al²⁶ found 70% of women postmenopause performed their exercises 2 to 3 times or more per week after 12 months. In 3 other longer-term studies (6–15 years fol-

lowing supervised intervention), only 5% to 12% of women reported doing exercises daily, and 50% to 70% performed them at any given time.^{16,27,28}

During the follow-up phase of this study, the only barrier significantly associated with decreased PFM exercise adherence was "difficulty finding time to do all the exercises." However, the impact of this barrier was low, with women performing only an average 2.5 fewer exercises per day compared with women who did not report this barrier. Therefore, other variables not investigated in this study may have been responsible for the decline in exercise adherence over time.

Alewijnse et al¹² reported incontinence severity and women's perception of their ability to exercise (self-efficacy) predicted adherence. Messer and colleagues²⁹ also identified self-efficacy regarding exercise technique and the ability to adhere to the PFM exercise regimen in the short term as predictors of long-term (12-month) PFM exercise. We did not query self-efficacy, but instead asked women whether they were "unsure if they were doing the exercises correctly." The ability to demonstrate a PFM contraction was an inclusion criterion for the study, and 96% of our sample contracted their PFM muscles without performing the Valsalva maneuver or using accessory muscles by the end of the intervention phase. Therefore, it is not surprising that we did not find uncertainty regarding exercise technique to predict exercise adherence.

We believe the lack of interventionist contact during the follow-up phase may have contributed to the decline in exercise adherence. Group support and regular monitoring by a health care professional have been identified as important contributors to long-term exercise in

studies of people with chronic obstructive pulmonary disease and of people following cardiac rehabilitation.^{30,31} Exercise reminders sent to patients by either postal or electronic mail may be an effective and more cost-efficient alternative for promoting long-term exercise adherence. Future studies should investigate the benefit of providing these types of contacts between the patient and health care provider to promote long-term PFM exercise adherence.

Another possibility to promote long-term adherence is to integrate PFM exercise into general fitness programs conducted at community centers or private health clubs. Brubaker et al³² designed and tested an 11-week fitness and education program aimed at promoting pelvic health in a sample of women with UI recruited from the community who were not actively seeking medical care. Women who completed the program reported improvement in urinary symptoms and quality of life. Outcomes were sustained or improved further 1 year following the program's completion.³² Whether this type of program could benefit women who previously underwent a behavioral intervention for UI to enhance long-term exercise outcomes is unknown and warrants investigation.

This study has several limitations. The exercise adherence questionnaire we designed for this study was not tested psychometrically. In addition, barriers were stated in question format with a simple 3-choice categorical response. We chose the yes/no/uncertain response scale to enable the interventionists to identify whether a particular barrier interfered with exercise. A Likert-type response scale may have provided greater information about the extent to which each reported barrier affected exercise adherence ("very

often,” “often,” “occasionally,” “rarely,” or “never”). Although the barrier questionnaire probed personal, environmental, and treatment-related variables, it was brief. Other barriers not measured (eg, self-efficacy or self-discipline regarding exercise) may have had an impact on exercise adherence. Therefore, our conclusions regarding predictors of PFM exercise adherence are limited.

Our study differed from others reported in the literature in that behavioral interventions were implemented together with antimuscarinic medication, which may have influenced the desire or perceived need to exercise. However, fewer than 1 in 5 women reported that the medication seemed more important than doing the exercises in the intervention phase.

Given the nature of a combined intervention, it is impossible to discern the actual impact of exercise adherence on study outcomes. At the end of the 10-week intervention period, 69% of the women experienced a $\geq 70\%$ reduction in UI episodes. These women performed on average of 31 PFM contractions per day, which did not differ from the average of 29 PFM contractions per day performed by women with less treatment success ($P=.39$). Thus, it may seem that exercise had no impact on outcomes during the intervention phase. However, it is possible that the drug intervention produced a differential effect on women that influenced outcomes despite the number of PFM contractions performed. In addition, exercise was only one component of the behavioral intervention. Ninety-seven percent of the women were able to perform a PFM contraction. All of the women were taught the urge suppression strategy, and 92% indicated they could use the strategy effectively to stop a UUI episode. Fewer women received stress strategy training, as it was based on

evidence of stress UI episodes in their bladder diary at week 2 of the intervention. Fifty-one percent of the women were taught this strategy, and 25% used it successfully by the end of the intervention period to prevent stress UI. Finally, 60% of the women achieved the ability to delay voiding. Thus, it is possible that once a woman developed enough PFM strength and skill to prevent urine leakage and delay voiding, the number of contractions needed to maintain these skills was significantly less than the average number of contractions these women actually performed or were asked to perform. It also is possible that once a woman became proficient and successful in using these strategies, PFM exercise was viewed as less important. Therefore, long-term adherence to and effectiveness of these strategies should be researched in future studies.

At the 10-week point, the medication was stopped. Fifty-six percent of the women achieved a $\geq 70\%$ reduction in UI episodes from baseline to the end of the 6-month follow-up. These women performed on average 9.2 PFM contractions per day, not different from the 8.9 PFM contractions per day performed by those who failed treatment ($P=.81$). If a woman requested to resume drug therapy, she was considered a study failure. Therefore, our follow-up exercise adherence data includes both women who resumed drug therapy and those who did not. As discussed previously, almost all of these women had achieved skill in using the urge strategy, and more than half were skilled in delayed voiding. Thus, it is difficult to conclude the impact of exercise adherence alone on our primary study outcome (the ability to discontinue drug therapy and achieve a $\geq 70\%$ reduction in UI episodes at the 8-month endpoint).

Strengths of this study include the number of data collection times dur-

ing the intervention and follow-up phases; the 12-month follow-up; the focus on measuring barriers to adherence; and given the number of participating sites and interventionists involved, the generalizability of the results.

Conclusions

During 10 weeks of active treatment, women with urge-predominant UI demonstrated high rates of adherence to behavioral interventions, including PFM exercises, urge suppression strategies, stress strategies, and delayed voiding. Adherence to PFM exercise was sustained at lower rates during the 12-month follow-up. Finding time to exercise was a barrier that predicted adherence during active intervention and throughout the 12-month follow-up. Associating exercise with other well-established behaviors or daily events may promote adherence by helping women remember to exercise. Other cost-effective means for promoting long-term PFM exercise, such as supervised group training, need to be investigated.

Dr Borello-France, Dr Burgio, Dr Goode, and Dr Stoddard provided concept/idea/research design. Dr Borello-France, Dr Burgio, Dr Goode, and Dr Markland provided writing. Dr Borello-France, Dr Burgio, Dr Goode, Ms Balasubramanyam, and Dr Stoddard provided data analysis. Dr Borello-France provided project management. Dr Goode provided participants and facilities/equipment. Dr Goode and Dr Kenton provided consultation (including review of manuscript before submission).

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Adherence to Behavioral Interventions for Urge Incontinence

Appendix 1.

Urinary Incontinence Treatment Network^a

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^a Asterisks indicate participating clinical centers.

Appendix 2.

Exercise Barrier Questions

- Was it difficult to find the time to do all of the exercises?
- Were you unsure if you were doing the exercises correctly?
- Did the exercises cause any pain?
- Did you have difficulty remembering the exercises?
- Did it seem as if the exercises were not helping your incontinence?
- Were you so much better that you felt that you did not need to continue the exercises?
- Did it seem that taking the medication was more important than doing the exercises? (asked at visits 2–4 only)
- Are there any other reasons you did not do the exercises?

Invited Commentary

Meena M. Sran, Penny Wilson, Pat Lieblich

The article by Borello-France and colleagues¹ addresses a critical issue in physical therapy: adherence to the prescribed exercise program. We commend the authors for their efforts to better understand barriers to adherence.

We have 7 primary comments and concerns about this study, important considerations for interpretation of their results, application in practice, and design of future studies.

All Participants Received Some Behavioral Component

The main purpose of the primary study, published in 2008, was to compare drug therapy in combination with a behavioral intervention for urinary incontinence versus drug therapy alone on discontinuation of drug treatment.² The primary study used a parallel group design, with all patients going off therapy at 10 weeks. A limitation of this study, acknowledged by the authors, was that both groups received some behavioral therapy, when in fact routine treatment with urinary incontinence medication does not include behavioral therapy. Knowing about the behavioral recommendation adherence of the group that received drug therapy only could shed some

light on the degree to which they relied on the behavioral components of their treatment and further our understanding of the results of this trial. Furthermore, it is not standard practice for patients to be advised to discontinue drug therapy after 10 weeks. We believe this is a critical limitation of this study and suggest that inclusion of a treatment group that received behavioral therapy alone would have provided more clinically relevant results.

Appropriateness of the Exercise Training Protocol—Lack of Overload?

In their systematic review, Neumann et al³ recommended that training programs should be adjusted for differences in initial strength (force-generating capacity) of the pelvic-floor muscles (PFMs) and symptom severity. In the study by Borello-France and colleagues, the PFMs were digitally assessed at each visit, but the exercise progression followed a set protocol regardless of the findings of the PFM assessment. For example, at visit 1, the maximum prescribed contraction was 3 seconds long, followed by a 6-second rest period. In the population they studied, we would expect variability in PFM strength, such that some par-

ticipants could easily perform a 3-second contraction at visit 1. The American College of Sports Medicine (ACSM) recommends progressively overloading skeletal muscle to increase strength and endurance.⁴ Thus, the protocol used by Borello-France et al may not have been adequate to provide overload of the PFMs for some participants.

Lengthen or Shorten the Rest Period to Overload Muscle?

Reducing the length of the rest period has been recommended as a method of overloading to increase muscle strength.⁵ By contrast, the recently published revised ACSM position stand on progressions in resistance training⁴ suggests that the rest period should be lengthened when training muscle for strength and shortened when training for endurance, and the report indicates there is a lack of clarity regarding rest period length when training for muscle hypertrophy. Borello-France and colleagues' PFM training protocol consistently used a rest time that was twice as long as the contraction time. We believe this approach highlights 2 important issues in PFM training that warrant further examination: